Stochastic Projective Embeddings of Dynamical System

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Abstract

Projective Embedding of Dynamical System (PEDS) is an algorithm proposed to find the global minimum for the non-convex loss functions [3]. It is inspired from the memristor circuits, which showcase certain capability to converge to the global minimum even if it's initialized outside of the basin of attraction of the global minimum [2]. To understand this, we connect PEDS with Particle Swarm Optimization (PSO) and argue that PEDS is a particle-based algorithm. Furthermore, we propose the large number limit of PEDS, called Sto-PEDS, if we increase the number of particles and treat the particles as samples from a distribution. Sto-PEDS turns out to have a nice analytical explanation for its efficacy. We demonstrate that Sto-PEDS is able to achieve better global minimum convergence rate and optimize to a lower loss than PEDS. It is promising that Sto-PEDS explains the efficacy in PEDS, and then the tunneling effect observed in memristor circuits, and possibly other natural systems.

1 Introduction

Gradient-based methods are frequently used in optimizations because they use both zero-th order and first-order information of the loss landscape. If the loss landscape is convex, gradient descent and newtons methods are known to converge to the global minimum with guarantee [1]. However, for the non-convex loss landscapes, the case is more subtle. There is no algorithm that perform strictly better than the brute-force algorithm (like grid-search) in all loss landscape. For example, imagine a landscape to be uniformly fat except at some point the landscape drops. So, people generally limit the function space for the loss function when discussing the efficiency of optimization algorithm.

It is discovered that memristor circuits showcase the ability to converge to the global minimum even when the initialization is out of the basins of attractions of the global minimum [2]. Memristors are a type of electronic device that evolves its internal state, which affects the resistance, according to the voltage. The authors conducted extensive experiments to show its convergence to the global minimum and used a mean-field approach to analyze the circuit dynamics [3], but the reason for global convergence is not clear. In a followup paper, an algorithm called Projective Embeddings of Dynamical System (PEDS) is proposed, largely inspired by the memristor circuits and it proves that PEDS transforms the loss landscape, turning local maximum to saddle points and preserving local

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minimum and saddle points [3]. This is intriguing, but we will show that this doesn't increase the probability of converging to the global minimum, compared to some naive approach with similar computations. Furthermore, we draw the connection between PEDS and Particle Swarm Optimization (PSO), demonstrating that PEDS can be viewed as a multi-agent optimization algorithm. Seeing the agents as samples from a distribution, we are inspired to send the number of agents to infinity and we propose a new algorithm called Sto-PEDS, which optimize the distribution directly. Through a large amount of experiments, we show that Sto-PEDS performs better and faster than PEDS. Moreover, Sto-PEDS has a better analytical explanation on why it works.

2 Preliminaries

2.1 Optimization

Suppose the function we want to optimize is $F(X)$ and the gradient is $f(X)$, i.e. $f(X) = \nabla F(X)$, where $X \in \mathbb{R}^m$ and F and f is a function from $\mathbb{R}^m$ to $\mathbb{R}$. Then gradient-descent can be formulated as an ODE by

$$\frac{dX}{dt} = -f(X). \tag{1}$$

2.2 Projective Embeddings of Dynamical System (PEDS)

To write out the formulation for PEDS, we need to first create N copies of X, calling them X_i for $i = 1, 2, \cdots, N$. Further, for $j = 1, 2, \cdots, m$ define $Y_j = (X_1^j, X_2^j, \cdots, X_N^j)$, where the super script denotes jth element for the vector. In other words, given a vector variable, we treat it as a column vector of the elevated matrix variable. Then, Y_j is the jth row and X_i is the ith column.

Next, the most general formulation for PEDS is

$$\frac{dY_j}{dt} = \Omega\Phi_f(\Omega; Y_1, Y_2, \cdots, Y_m) + G(\Omega; Y_j), \tag{2}$$

where Φ_f is a function that relates the gradient function f, G is a decay function, and Ω is a projector (that is, $\Omega^2 = \Omega$). We defer the specific meaning of these terms to the original paper [3]. In the original paper, they analyzed the case when Ω is a mean-field projector, meaning it's a matrix filled with $1/N$. With some simple choice of Φ and G, the update for PEDS becomes

$$\frac{dY_j}{dt} = \Omega_1 \vec{f}(Y_1, Y_2, \cdots, Y_m) - \alpha(I - \Omega_1)Y_j, \tag{3}$$

where $\vec{f}^k = f(Y_1^k, Y_2^k, \cdots, Y_m^k)$, Ω_1 is the mean-field projector (matrix with $1/N$), and α is a hyperparameter.

2.3 Particle Swarm Optimization

With the strange formulation in Eq. 3, we can rewrite this formulation in terms of X_i, which gives us

$$\frac{dX_i}{dt} = \frac{1}{N}\sum_{k=1}^{N} f(X_k) - \alpha(X_i - \frac{1}{N}\sum_{k=1}^{N} X_k). \tag{4}$$

We interpret this intuitively. Every X_i is an agent optimizing on the loss landscape. Each agent X_i will sense the gradients of other agents, as looking at the first term, it is the average of the gradients of all agents. Moreover, each agent will have a tendency to converge to each other, as looking at the second term, it essentially pulls the current agent to the collective center of all agents. The hyperparameter α determines how strong is the attraction.

This is different from the traditional view of Particle Swarm Optimization (PSO), which sets the maximum value of the particles as an anchor point and pull particles towards that point gradually [5]. Though the original doesn't involve the gradients, there has been a lot of work extending PSO with gradient information [8]. There are also other generalizations in PSO, like investigating the division of subpopulations, combining Nelder-Mean simplex search with PSO, and using the inertial weight

to constrain the step size when close to the global minimum [6], [4], [9]. There is also extension which transfers the weights between particles depending on the loss function while doing gradient descents[7]. Our work views the center of the particle swarm as our target optimization value. This is different to the general idea of PSO.

3 Method

3.1 Sto-PEDS

Inspired by the formulation in Eq. 3, we design so-called Stochastic Projective Embeddings of Dynamical Systems (Sto-PEDS). To see this, observe that the first term in Eq. 3 is an empirical approximation for $\mathbb{E}_X f(X)$ where X_i are samples from the some distribution. For the sake of simplicity, let us assume the distribution is Gaussian. Then, $\frac{1}{N}\sum_{k=1}^{N} f(X_k) \approx \mathbb{E}_{X\sim\mathcal{N}(\theta,\sigma I)}$. For the second term, all the particles are pulled closer to the center of mass, that is, the mean of the distribution. This is equivalent to shrink the variance of the distribution. Hence, Sto-PEDS is proposed. Further, we make some remarks about Eq. 3. First, if we set α to be 0, the shape of X_k will not change because every agent has the same update rule. This is equivalent to optimize the mean of the distribution. Second, if we set α to be very large, the particles will just converge to its center, without using too much information from f at all. These two remarks justify our interpretation for the distribution. The first term is optimizing the mean and the second term is decreasing the variance. Combining these two steps together, we propose Sto-PEDS, which is described in details in Alg .1

Algorithm 1 Stochastic Projective Embeddings of Dynamical System (Sto-PEDS)

Require: γ: Stepsize
Require: N: Number of samples
Require: $f(X)$: Gradient function
Require: α: Linear decreasing rate for standard deviation
Require: θ_0, σ_0: Initial parameters
$t \leftarrow 0$ (Initialize timestep)
while σ_t is not small **do**
  $t \leftarrow t + 1$
  $R_1, \ldots, R_N \sim \mathcal{N}(\theta_{t-1}, \sigma_{t-1} I)$ (Draw samples from the normal distribution)
  $g_i \leftarrow f(R_i)$ (Compute the gradient for each sample)
  $g_t \leftarrow \frac{1}{N}\sum_{i=1}^{N} g_i$ (Compute the average)
  $\theta_t \leftarrow \theta_t - \gamma * g_t$ (Update the parameter)
  $\sigma_t \leftarrow \max(\sigma_{t-1} - \alpha, 0)$ (Update the std)
end while
return θ_t (Mean of the resulting distribution)

3.2 Modified PEDS

One difference we should notice about standard PEDS and Sto-PEDS is that according to Eq. 3, the distance of the agents decreases exponentially whereas the variance of Sto-PEDS decreases linearly by its definition. To take away this difference, we implement a slightly difference regime for PEDS so that the distance between particles decrease linearly. This is achieved to set α to be proportional to $\log$ of the average distance to the center of mass.

One more remark for this. For the Sto-PEDS, the number of iteration is known because the variance is decreasing at a constant rate and the stopping condition is when the variance reaches zero. For PEDS, it is the same unless that the random initialization gives the initial distance some variability, so its number of iteration with different random initialization will differ slightly. This is helpful for us to compare their efficacy because we can easily control their computation requirements.

3.3 Fourier Analysis for Sto-PEDS

For Sto-PEDS, we can analyze it with Fourier analysis. In particular, the first step in Sto-PEDS is to approximate

$$\nabla \mathbb{E}F(R) = \mathbb{E}\nabla F(R) = \int \nabla F(R)\mathcal{N}(R;\theta,\sigma I)dR = \int \nabla F(R)\rho(\theta - R)dX = \nabla F * \rho(\theta), \quad (5)$$

where $\rho(X) \approx e^{-\|X\|^2}$. Taking the Fourier transform on both sides, we get

$$\mathcal{F}\{\nabla \mathbb{E}F(R)\} = \mathcal{F}\{\nabla F\}\ \mathcal{F}\{\rho(\theta)\}, \quad (6)$$

where $\rho(\theta)$ will be a low pass filter for ∇F in the frequency domain. Moreover, higher σ means only signals with lower frequency can pass. Therefore, if the high frequency signal of ∇F is noise, Sto-PEDS can filter out them and keep the low frequency signal. And as time goes by, σ decreases, there will be more high frequency signal available for the algorithm to optimize.

4 Experiments

To show the efficacy of PEDS and Sto-PEDS, we conduct experiments on various hyper-parameters. We choose Ackley and Rastrigin on our test functions. We have two metrics for our experiments. The first one is success rate, which means whether the algorithm finds the global minimum. The second one is loss, which is the function value when the algorithm stops. The hyperparameters for the test functions are chosen that the global minimum has value 0.

4.1 Success rate

First, the success rate is plotted in Figure. 1. Sto-PEDS showcases a better success rate than PEDS on rastrigin and their performance on ackley functions are almost the same.

4.2 Average loss

If we look at the success rate for Sto-PEDS, we notice that it happens that in some case, experiments with larger m have a larger success rate (see in Figure. 2). When we plot the average loss, we figure out that the loss is actually pretty low. The loss showcases that the algorithm is fairly closed to the global minimum, but it cannot find the exact position because of the stochastic nature of Sto-PEDS. See that in Figure. 5, the blue line has a rising tail at the end of the convergence. In the case where m is slightly larger, the landscape provides stronger signal because of the dimensions, so the uncertainty brought by the noise is less significant. In practice, we can concatenate a convex optimization algorithm like Newton's method after Sto-PEDS to find the exact global minimum. As long as Sto-PEDS brings the point to the basin of attractions of the global minimum, convex optimization is guaranteed to find the global minimum.

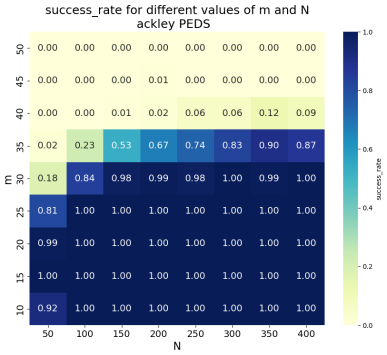
Then, we look at the average loss. The result is plotted in Figure. 3. Sto-PEDS results in a much lower loss than PEDS.

4.3 Variance reduction rate

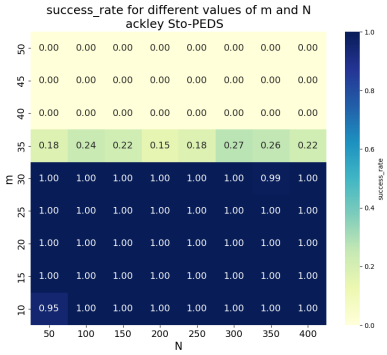
One important observation from the experiments is that success rates highly depend on variance reduction rate (how fast we decrease the variance). Smaller variance reduction rate implies more computations. The result is shown in Figure 4. In the previous sections, we simply use the success rate or average loss of the best variance reduction rate. It is possible that with lower variance reduction rate, there will be different behaviors, but the computation is inversely proportional to the value of variance reduction rate, so it is not worthwhile to try that.

4.4 Annealing

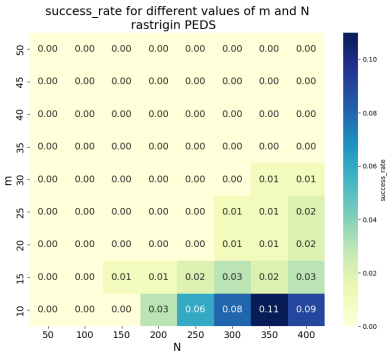
A natural question to ask is that if it is possible to find the variance reduction rate adaptively, so it is less sensitive to the hyperparameter setting. Here, we proposed an approach for annealing the variance reduction rate. The idea is that if the algorithm is moving faster (the average norm of the gradient), this means the algorithm is exploiting from the potential, so increase the variance. On the other hand, if it is moving slower, decrease the variance. Because the average steps may be noisy, we



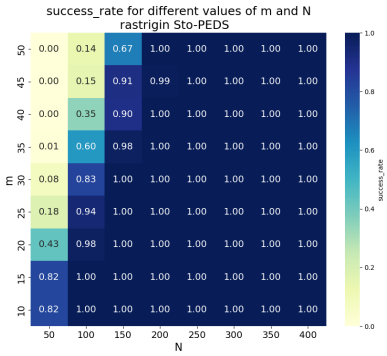
(a) PEDS on Ackley functions



(b) Sto-PEDS on Ackley functions

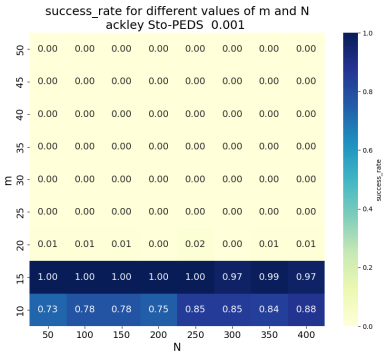


(c) PEDS on Rastrigin functions

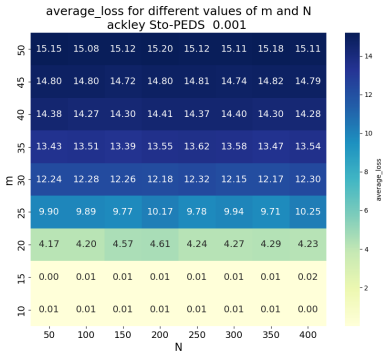


(d) Sto-PEDS on Rastrigin functions

Figure 1. Success rate for PEDS and Sto-PEDS on Ackley and Rastrigin test functions. Observe that Sto-PEDS perform better than PEDS on rastrigin functions and they have comparable performance for Ackley functions.

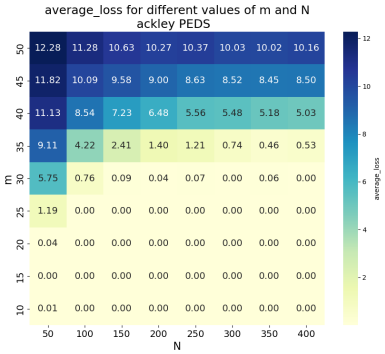


(a) Success rate. $m = 10$ has a lower success rate than $m = 15$ with the same N. This is uncommon. Larger m should have lower success rate because the problem has higher dimension.

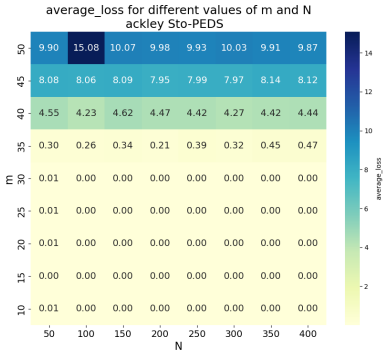


(b) Average loss. For $m = 10$ and $m = 15$, their average loss are comparable and pretty low. This shows consistency with our expectation that higher m implies more difficulty.

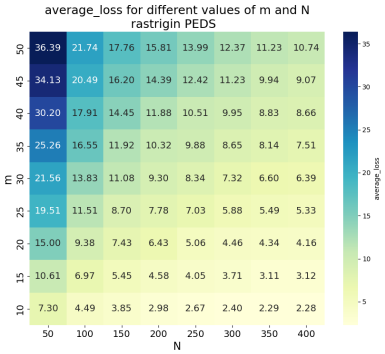
Figure 2. Success rate inversion



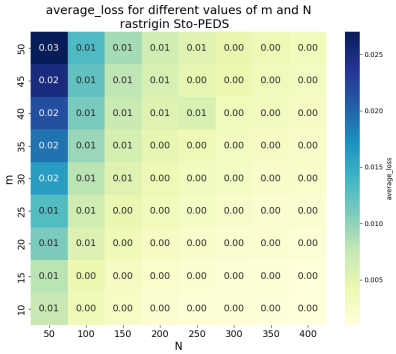
(a) PEDS on Ackley functions



(b) Sto-PEDS on Ackley functions

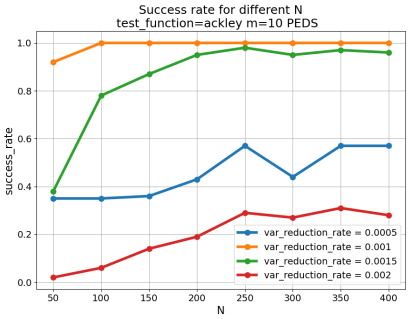


(c) PEDS on Rastrigin functions

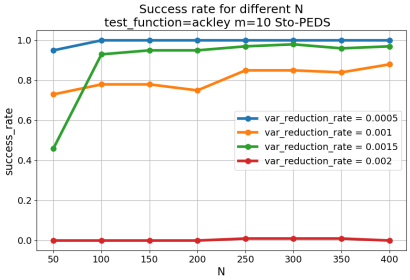


(d) Sto-PEDS on Rastrigin functions

Figure 3. Average loss for PEDS and Sto-PEDS on Ackley and Rastrigin test functions. Observe that Sto-PEDS perform better than PEDS on both ackley and rastrigin functions. In cases where the global minimum is not found, Sto-PEDS is closer to the global minimum than PEDS.



(a) PEDS



(b) Sto-PEDS

Figure 4. Success rate for PEDS and Sto-PEDS on Ackley test functions and $m = 10$ for different variance reduction rates. For both algorithms, variance reduction rate affects the performance of the algorithm. Smaller variance reduction rate generally implies higher success rate.

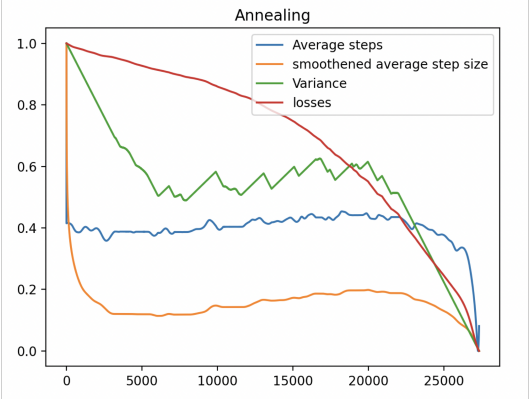
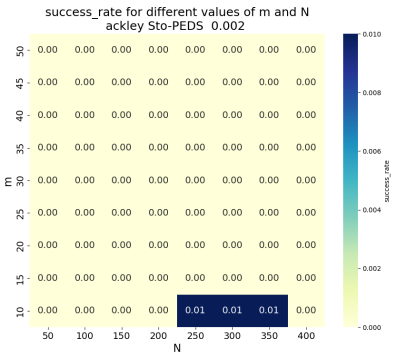


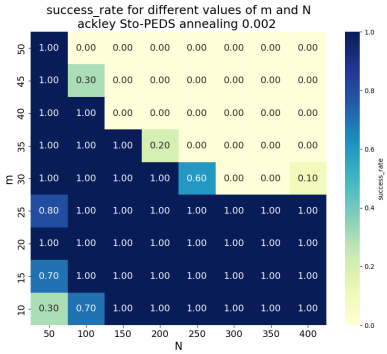
Figure 5. Annealing illustration. All the quantities are normalized for illustration. The orange line is the smoothened version of the blue line, and it is used to determine the variance, which is the green line. When the orange line going upwards, the green line also goes upward. Otherwise, the green line will go downwards.

use a moving average as the criterion. Also, to facilitate the convergence, the rate of variance increase is half of the rate of variance decrease. The annealing process is illustrated in Figure. 5.

With annealing, we show that the convergence is less sensitive to the setting of variance reduction rate. In Figure. 6, the success rate increases significantly from the usage of annealing, though the annealing scheme takes strictly more time to converge. There is probably more sophisticated scheme to anneal the variance reduction rate, which is left for future work.



(a) Without annealing. The success rate is almost 0 because the variance reduction is too large. The contraction contracts to the single point too fast that it doesn't have time to explore the landscape.



(b) With annealing. The performance is much better. There are two regions where success rate is not 1. Increasing N will increase the mollyfing effect of Sto-PEDS, but it also adds noise to the method. The lower-left corner doesn't succeed. Either adding more signal by increasing m or decreasing the noise by increasing N can solve the problem. In the upper-right corner, larger N is harmful because it brings too much noise.

Figure 6. Success rate inversion

4.5 One agent case

It is interesting that setting N to be 1, Sto-PEDS can still converge. It is not surprising because it is an unbiased estimator of the expectation of the gradient, so it should converge, despite of the variance, in this naive loss landscape.

5 Conclusion

PEDS is an algorithm inspired by memristor circuits. Because of its complicated structure, analyzing it is difficult. Previous works prove that PEDS convert local maximima to saddle points [3], but the reason of converging to the global minimum is still unclear. In this work, we connect PEDS with the idea of Sto-PEDS, by considering a particular case of PEDS. Sto-PEDS has a straightforward analytical formulation, so it is clear that it works on some hypothesis set of loss functions, but no the others. Through experiments, we showcased that Sto-PEDS performs better than PEDS, in terms of the probability of converging to the global minimum and the average loss after optimization. It is strongly believed that Sto-PEDS provides a potential explanation for the efficacy of PEDS.

In the experiments, however, the loss function is primitive and analytically Sto-PEDS should work perfectly. The more interesting problem is whether Sto-PEDS performs well on other natural systems, like the Lennard-jones cluster of 13 particles. This will be the next step of our work.

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